



**DELHI PUBLIC SCHOOL NEWTOWN**  
**SESSION: 2024-25**  
**HALF YEARLY EXAMINATION**

**CLASS: X**  
**SUBJECT: COMPUTER APPLICATIONS [SET A]**

**FULL MARKS: 100**  
**TIME: 2 HOURS**

*Answers to this Paper must be written on the paper provided separately.*

*You will **not** be allowed to write during the first **15** minutes.*

*This time is to be spent in reading the question paper.*

*The time given at the head of this Paper is the time allowed for writing the answers.*

*This paper consist of ten printed pages. This Paper is divided into **two** Sections.*

*Attempt **all** questions from **Section A** and **any four** questions from **Section B**.*

*The intended marks for questions or parts of questions are given in brackets[ ].*

**SECTION A**

*(Attempt **all** questions from this **Section**)*

**Question 1**

Choose the correct answer and write the correct option.

[20x1=20]

(i) The \_\_\_\_\_ method is used to convert **String** data to **float** type.

- |                          |                           |
|--------------------------|---------------------------|
| (a) parseFloat(String s) | (b) parseFloat(int s)     |
| (c) parseString(int s)   | (d) parseString(String s) |

(ii) Choose the correct explanation of **Math.random( )** function:

- (a) return a pseudorandom integer type number greater than or equal to 0.0 and less than 1.0
- (b) return a pseudorandom double type number greater than or equal to 0.0 and less than 1.0
- (c) return a pseudorandom float type number greater than or equal to 0.0 and less than 1.0
- (d) return a pseudorandom integer type number lesser than or equal to 0.0 and more than 1.0

(iii) Which of the following method **do not** return a **boolean** data type:

- |                               |                               |
|-------------------------------|-------------------------------|
| (a) Character.isLowerCase(ch) | (b) Character.isUpperCase(ch) |
| (c) Character.isDigit(ch)     | (d) Character.toLowerCase(ch) |

(iv) Predict the output: **System.out.println(Math.floor(-3.5+ Math.ceil( - 0.7)));**

- |          |         |          |        |
|----------|---------|----------|--------|
| (a) -4.0 | (b) 3.0 | (c) -3.0 | (d) -4 |
|----------|---------|----------|--------|

(v) If `int x[] = { 4, 3, 7, 8, 9, 10 };` What will be the value of 'p' and 'q'?

`p = x.length;`                      `q = x[2] + x[5] * x[1];`

- (a) 5, 36                      (b) 6, 36                      (c) 6, 37                      (d) 5, 37

(vi) Evaluate the expressions: `Q = ( x-- % 5 ==5) &&(y / 2 > x* --y)` , where, `x = 0` and `y = -7`

- (a) true                      (b) false                      (c) error                      (d) no output

(vii) \_\_\_\_\_ method does not return the same value every time it is called, even when the same arguments are given whereas \_\_\_\_\_ method always gives the same result when the same arguments are passed.

- (a) Pure, impure                      (b) impure, pure  
(c) static, non static                      (d) non static, static

(viii) Which of the following is applicable for a constructor function compared to methods?

- (a) It has the same name as the class with no return type.  
(b) It has no return-type and no arguments  
(c) It is not automatically called when object is created. Needs to be invoked in `main()`.  
(d) It can have different names with a specific return type and arguments

(ix) How many **times** loop is executed in the given program:

```
int a, b;  
for(a=6, b=4; a<=24; a=a+6)  
{ if(a%b==0) break; }  
System.out.println(a);
```

- (a) 3                      (b) 2                      (c) 1                      (d) 0

(x) Predict the output:

```
public class chk  
{ chk(char a)  
    { System.out.println("Value="+ a ); }  
  chk(double a)  
    { System.out.println("Value=" + a); }  
}
```

```
public static void main()
{   chk c1 = new chk('h');
    chk c2 = new chk(49); } }
```

- (a) Value=h  
Value=49.0
- (b) Value=h  
Value=49
- (c) Compiler error
- (d) Semantic error

(xi) The element in **x[4]** of the array { 3, 5, 7, 12, 16, 18, 20, 25, 42, 89 } is

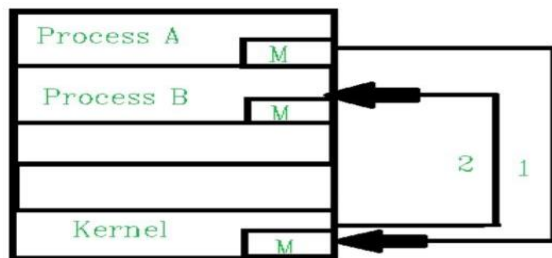
- (a) 16
- (b) 12
- (c) 7
- (d) 18

(xii) **Assertion(A):** In java, we use arrays to group similar type of data items..

**Reason(R):** Elements of the array are stored in continuous memory location.

- (a) Both **Assertion(A)** and **Reason(R)** are true and **(R)** is a correct explanation of **(A)**
- (b) Both **Assertion(A)** and **Reason(R)** are true and **(R)** is not a correct explanation of **(A)**
- (c) **Assertion(A)** is true and **Reason(R)** is false
- (d) **Assertion(A)** is false and **Reason(R)** is true

(xiii) Which **concept of java** is illustrated from the given image:



- (a) abstraction
- (b) Message passing
- (c) encapsulation
- (d) All three

(xiv) If **String s[] = { “Artificial Intelligence”, “IOI”, “Big Data”, “JDK” }**, then predict the

output: **System.out.println(x[3].length);**

- (a) 3
- (b) 2
- (c) error
- (d) 8

(xv) Which of the following statement will return the difference between the **next higher integer value** and **previous lower integer value**.

- (a) `System.out.print(Math.ceil( ) – Math.max( ))`
- (b) `System.out.print(Math.ceil( ) – Math.floor( ))`
- (c) `System.out.print(Math.floor( ) – Math.min( ))`
- (d) `System.out.print(Math.floor( ) – Math.ceil( ))`

(xvi) **Assertion(A)**: Each overloaded constructor must have a different prototype.

**Reason(R)**: Java compiler uses parameter lists, parameter types and number of input parameters to identify overloaded functions.

- (a) Both **Assertion(A)** and **Reason(R)** are true and **(R)** is a correct explanation of **(A)**
- (b) Both **Assertion(A)** and **Reason(R)** are true and **(R)** is not a correct explanation of **(A)**
- (c) **Assertion(A)** is true and **Reason(R)** is false
- (d) **Assertion(A)** is false and **Reason(R)** is true

(xvii) Predict the output:

```
class abc
{ static void main()
{ int n = 47389, d;
  while(n > 10)
  { d = n % 10;
    System.out.print(d);
    n = n / 100; }
  System.out.print(n); } }
```

- (a) 937
- (b) 983
- (c) 943
- (d) 934

(xviii) Which of this statement is **incorrect**?

- 1) switch statement is more efficient than a set of nested ifs
- 2) two case constants in the same switch can have identical values
- 3) switch statement can only test for equality, whereas if statement can evaluate any type of boolean expression
- 4) it is possible to create a nested switch statements

- (a) 1, 2
- (b) 3, 4
- (c) only 2
- (d) only 3

(xix) Real life example of **class** and **object** is:

(a) flower → rose, lotus, lily

(b) red rose, white rose, yellow rose

(c) both a and d

(d) herbivore → cow, goat, buffalo

(xx) The array **char x[10]** occupies:

(a) 10 bytes

(b) 40 bytes

(c) 20 bytes

(d) 80 bytes

## Question 2

(i) Predict the output:

[2]

```
char c[] = { 'm', 'n', 'o', 'p', 'q' };
```

```
int x=4;
```

```
System.out.println(c[x++]);
```

```
System.out.println(c[x]++);
```

(ii) How are **primitive** data types and **wrapper** classes different? Discuss with an example.

[2]

(iii) Diksha got an assignment from her school teacher to create a pattern using nested loops in java. She wrote the program but was not getting the desired output. What changes she should incorporate in the given program to reach the correct output?

[2]

**Assignment: A**

A C

A C E

Incorrect program:

```
public class PatternEx {  
    public static void main(String[] args)  
    { System.out.println("Displaying alphabet pattern: ");  
      for(int i = 65; i <= 69; i++)  
      { for(int j = 65; j <= i; j+=2) {  
        int ch = (char)j;  
        System.out.print( j+ " ");    }  
        System.out.print(" ");    } } }
```

(iv) In Java, the class forms the basis of all computation. A class is a blueprint for an object and an object is an instance of a class. It allows a programmer to define all of the properties and methods that define an object. Anything that has to exist as a part of a Java program has to exist as a part of a class, whether that is a variable or a function or any other code-fragment. Without classes, there can be no objects and without objects, no computation can take place in Java. Thus, classes form the basis of all computation in Java. Some of the components/features of a method are as described below. Write the appropriate term used for each component/ feature described above. [4]

- (a) It is defined as the process of creating objects of a class.
- (b) It is the identifiable entity that has a set of attributes, behavior and state.
- (c) It is a type of variable created when an object of the class is created and destroyed when the object is destroyed.
- (d) It is a type of variable created at the start of program execution and destroyed automatically when the execution ends.

(v) The following program is checking if the entered number is an **Automorphic number** or **not**. [4]

A number is called an **automorphic** number if and only if the **square** of the given number **ends** with the **same** number itself. For example, **25, 76 are automorphic numbers** because their square is 625 and 5776, respectively and the last two digits of the square represent the number itself. Substitute the numbered places with suitable java statement(s)/expression so that the program runs.

```
import java.util.Scanner;

public class AutomorphicNumber

{ public static void main( )

{ Scanner in = new Scanner(System.in);

System.out.print("Enter a number to check");

int num = in.nextInt(); int count=0;

int square = num*num; int temp = num;

while(__?1?__)

{ count++;
```

```

        __?2?__=temp/10; }

        int lastDigit = (int) (__?3?__ %(Math.pow(10, count)));

        if(num == __?4?__)

            System.out.println(num+ "is an automorphic number");

        else

            System.out.println(num+ " is not an automorphic number"); } }

```

(vi) Predict output:

[2]

```

class ede

{ void main( )

{ int i;

for( i=5 ; i>=1 ;i--)

{ if(i%2 ==1)

continue;

else

System.out.print(i--);

System.out.print( i+ " "); } } }

```

(vii) Rewrite the given statement using while loop:

[2]

```

for( int n=1, int s=0; n<=10; n=n+1)

s+=n;

```

(viii) Write the equivalent java expression:  $\frac{4 - \sqrt{|bc| - c^2}}{10a}$

[2]

## Section B

(Answer **any four** questions from this **Section**.)

*The answers in this section should consist of the programs in either BlueJ environment or any program environment with java as the base.*

*Each program should be written using variable description/mnemonic codes so that the logic of the program is clearly depicted.*

*Flowcharts and algorithms are not required.*

### Question 3

[15]

Define a class to declare a character array of size ten. Accept the characters into the array and display the characters with highest and lowest ASCII (American Standard Code for Information Interchange) value. Find the number of vowels and consonants. Finally display the original array in toggled case. (Toggle case changes upper case letters to lower case and vice versa)

EXAMPLE :

INPUT:

'R', 'z', 'q', 'A', 'N', 'p', 'm', 'U', 'Q', 'F'

OUTPUT :

Character with highest ASCII value = z

Character with lowest ASCII value = A

Number of vowels = 2

Number of consonants = 8

Toggled array = 'r', 'Z', 'Q', 'a', 'n', 'P', 'M', 'u', 'q', 'f'

### Question 4

[15]

Write a program to input a **4x4 array** and display the **sum** of its **left** and **right diagonals**.

### Question 5

[15]

Define a class to accept values into a **double** data type array of **size 20** and print the range of the array. Range is defined as the difference between the largest and smallest element of the array when arranged in ascending order. (Note: Use Bubble Sort technique). Also display the **ordered** array.

### Question 6

[15]

ISBN is another special number in Java. **ISBN** stands for the International Standard Book Number that is carried by almost each every book. The ISBN is a **ten-digit unique number**. With the help of the ISBN, we can easily find any book.



The ISBN number is a legal number when  $1*Digit1 + 2*Digit2 + 3*Digit3 + 4*Digit4 + 5*Digit5 + 6*Digit6 + 7*Digit7 + 8*Digit8 + 9*Digit9 + 10*Digit10$  is divisible by 11. The digits are taken from right to left. So, if the ten-digit number is 7426985414, Digit1 and Digit10 will be 4 and 7, respectively.

Let's take two numbers and check whether the numbers are legal or not.

**Example 1: 8147852369**

$$\text{Sum} = (1*9) + (2*6) + (3*3) + (4*2) + (5*5) + (6*8) + (7*7) + (8*4) + (9*1) + (10*8) = 281$$

Divide the sum with 11 and check whether the remainder is 0 or not.

If the remainder is 0, the number is a legal ISBN.  $\text{rem} = 281 \% 11 \rightarrow \text{rem} = 6 \neq 0$ ,

Number 8147852369 is not a legal ISBN

**Example 2: 1259060977**

$$\text{Sum} = (1*10) + (2*9) + (5*8) + (9*7) + (0*6) + (6*5) + (0*4) + (9*3) + (7*2) + (7*1) = 209$$

Now, we divide the sum with 11 and check whether the remainder is 0 or not.

$$\text{rem} = 209 \% 11 \rightarrow \text{rem} = 0$$

Number 1259060977 is a legal ISBN

**Question 7**

[15]

Design a class to **overload** a function **ser( )** as follows:

(a) **void ser(int a, int n)** — To display the series given below:

$$a - (a/2!) + (a/3!) - (a/4!) + \dots n \text{ terms}$$

(b) **double ser(int n)** — To display the sum of the series given below:

$$S = 1/2 - 2/3 + 3/4 - 4/5 + \dots n \text{ terms}$$

(c) **void ser( )** — to display the following pattern:

A B C D E

b c d e

C D E

d e

E

Write a **main** method to create an object and invoke the above methods.

**Question 8**

[15]

A company pays commission to its salesmen on the sale value according to the following rates:

| <b>Sales</b>   | <b>Commission</b> |
|----------------|-------------------|
| Upto 1000      | 5% on sale        |
| Next 1000      | 8% on sale        |
| Next 2000      | 12% on sale       |
| 4001 and above | 15% on sale       |

Design a program with the following specifications:

Class name: **Salecom**

Data members: sale, com

Member methods:

**Salecom( )**: default constructor

**Salecom( double s )** : constructor to initialize sale and com with 's' and 0 respectively

**void commission( )**: to calculate the commission based on mentioned rates

**void display( )**: to display the commission

Also write the **main( )** method to implement the above methods.